SOURCE CODE ARCHITECTURE

PennyPal

Make Every Penny Count.

PLATFORM	SCALE	VERSION
Flutter (Dart 3.5+)	161 Files, 20k+ Lines	0.1.0+1

CORE MISSION

Intelligent Student Ecosystem

Tracking & Budgeting

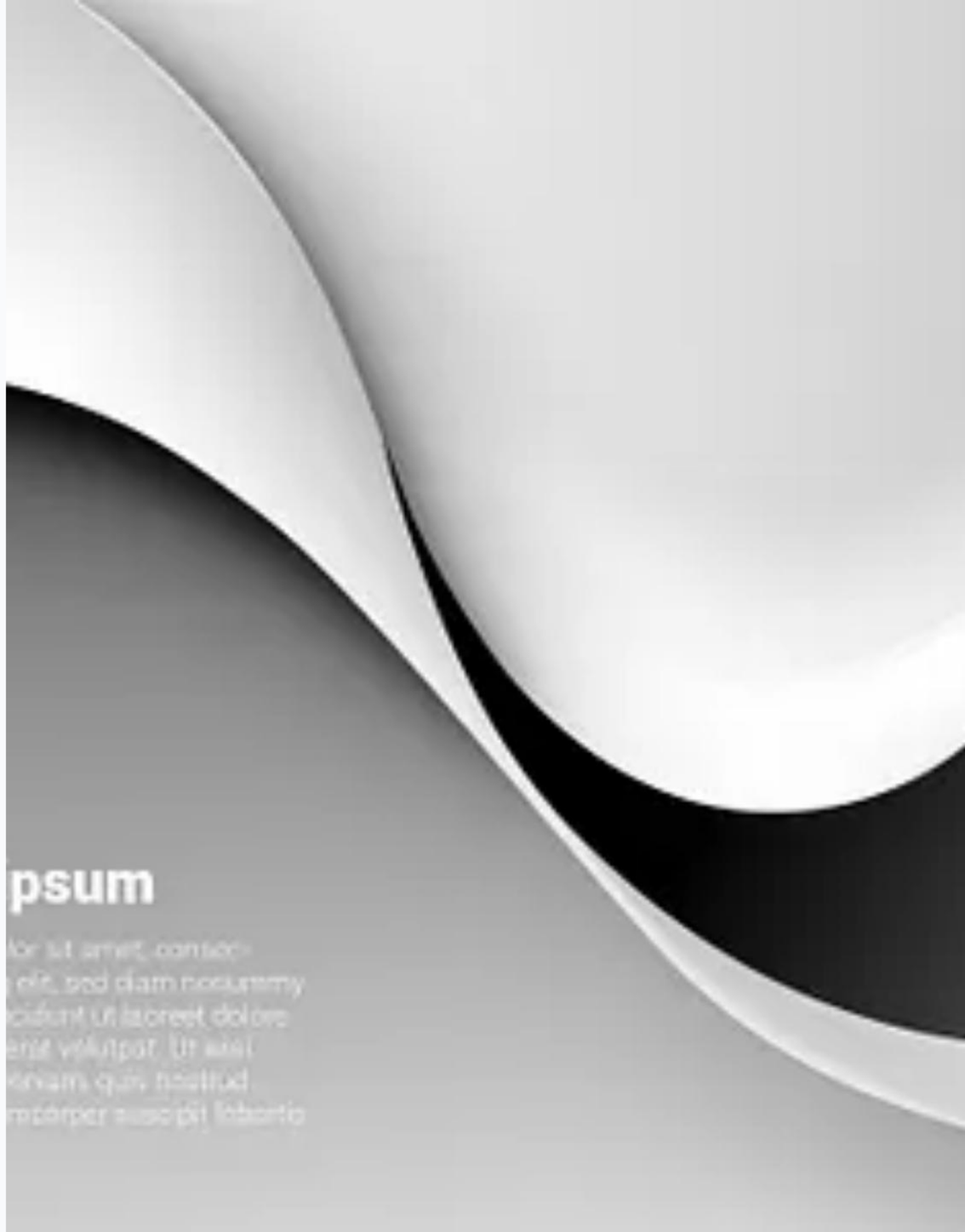
Automated expense monitoring and strict limit enforcement.

Goal Optimization

Milestone tracking mapping progress to 100% completion.

AI & Learning

Gemini-powered insights and financial literacy modules.



Technology Foundation

Modern dependencies powering the application shell.



Flutter Engine

Cross-platform UI.



Firebase

Auth and Firestore cloud.



Riverpod

Reactive state management.



Gemini AI

Intelligent assistant core.

MACRO STRUCTURE

5-Layer Application Architecture

1. Presentation — Screens, Sheets, Shared Widgets.

2. State (Riverpod) — Providers, Notifiers, Streams.

3. Domain
Pure Calculators & Models.

4. Data
Repositories & RecordStore.

5. Core Services — External API Wrappers.

CODE ORGANIZATION

Feature-First Design

Isolated modules containing unique logic, UI, and data components.

domain/

- Immutable models.
- toMap/fromRecord methods.
- Pure Dart calculators.

data/

- Repository classes.
- Riverpod providers.
- RecordStore connectivity.

presentation/

- ConsumerWidget screens.
- Bottom sheets and dialogs.
- State controllers.

Riverpod State

StreamProvider

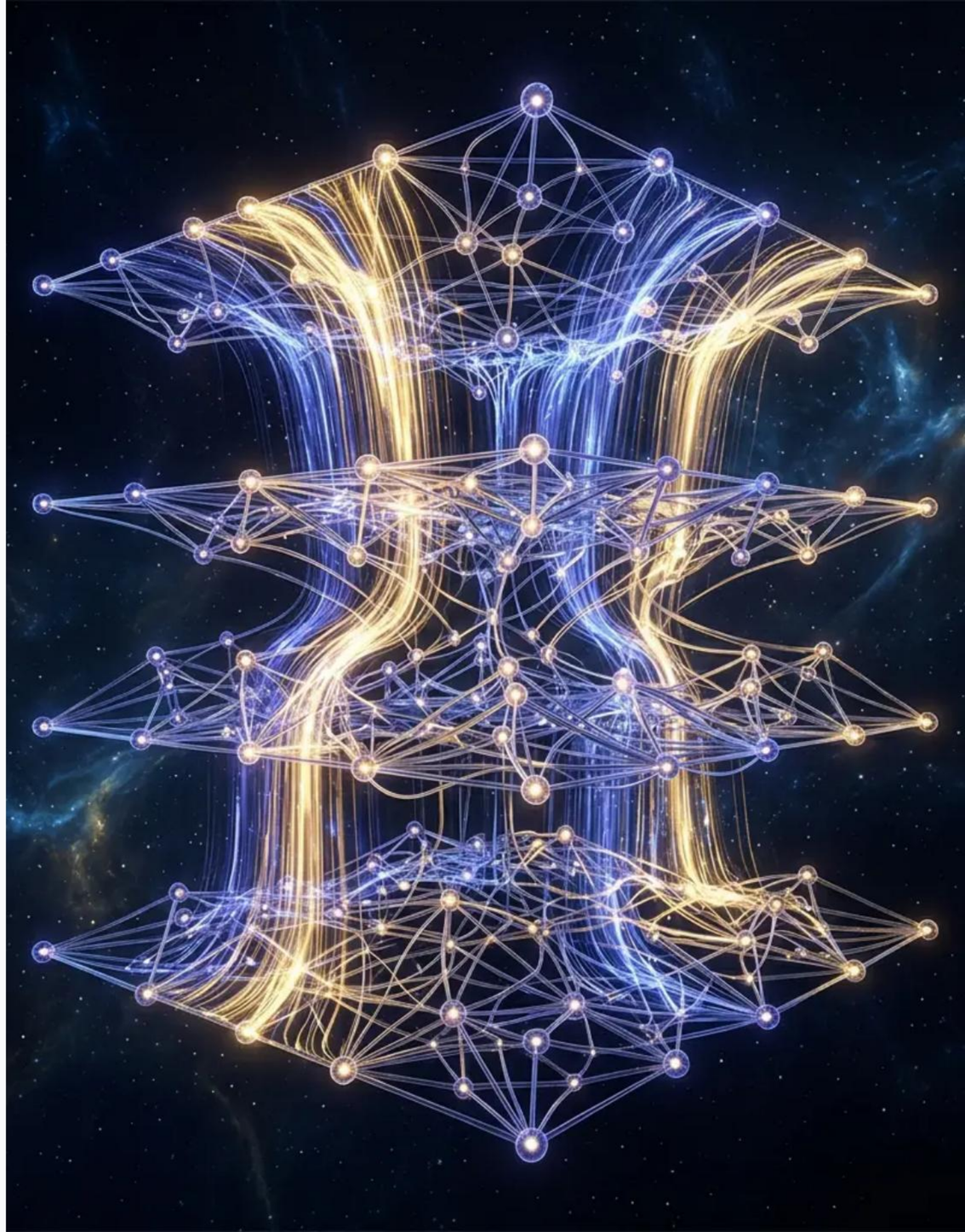
Live lists from Firestore syncing automatically.

NotifierProvider

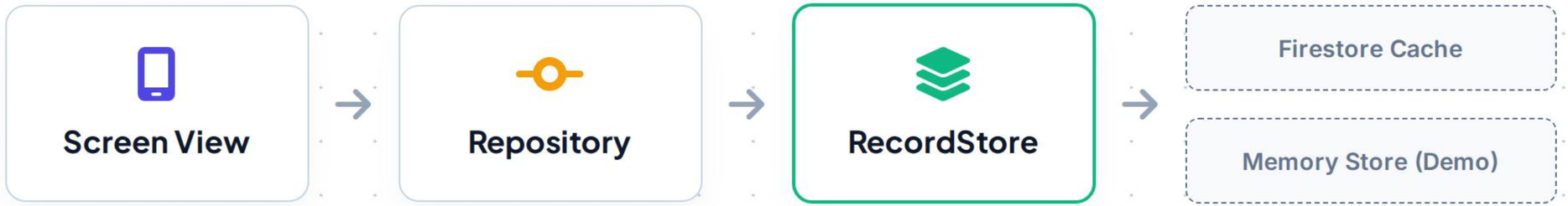
UI state management spanning multiple screens.

Derived Providers

Dependent screens automatically recalculate metrics.



Offline-First Persistence Flow



ROUTER ARCHITECTURE

Adaptive Navigation System



GoRouter Map

Declarative routing tables.



Redirect Guard

Single function enforces role access.



Adaptive Shells

Responsive rail and sidebar UI.

Authentication Pipeline



Register

Firebase user creation.

Send OTP

Brevo / Resend delivery.

Verify

Check 6-digit code validity.

Student Shell

Redirect to application.



FIRESTORE MODEL

Document Sub-collections

- **users/{uid}/transactions**
  Immutable financial records.
- **users/{uid}/budgets**
  Monthly category threshold caps.
- **users/{uid}/savingsGoals**
  Target tracking configurations.

Financial Health Index



Budget Pace

Spend vs. expected timeline.

Savings Ratio

Net surplus measurement.

Consistency

Daily logging evaluation.

Pure Domain Logic

Testable Dart functions absent of UI or Database dependencies.

Finance Calculator

Balance and mode summaries.

Budget Calculator

Status and ratio metrics.

Savings Calculator

Milestone projection logic.

Report Calculator

Period range comparisons.

Intelligent Transaction Logging

Expense Categorizer

Keyword-based tag suggestions.

Receipt OCR

Gemini vision amount parsing.



Transaction Record

Type	**Expense**
Category	**Transport**
Alerts	**Budget Threshold**

Budget Monitoring

Overall Month Limit 85%

Food & Dining 100%

Savings Goals



Deterministic milestone tracking triggers achievement alerts dynamically.

Gemini Assistant Core

Context-aware conversational interface with graceful offline degradation.

AssistantContext Injection
Feeds live budget and goal states to prompt.

250 Token Limit Constraint
Enforces student-friendly brief responses.

Rule-Based Fallback
Keyword matching engine during offline states.



Financial Literacy Modules

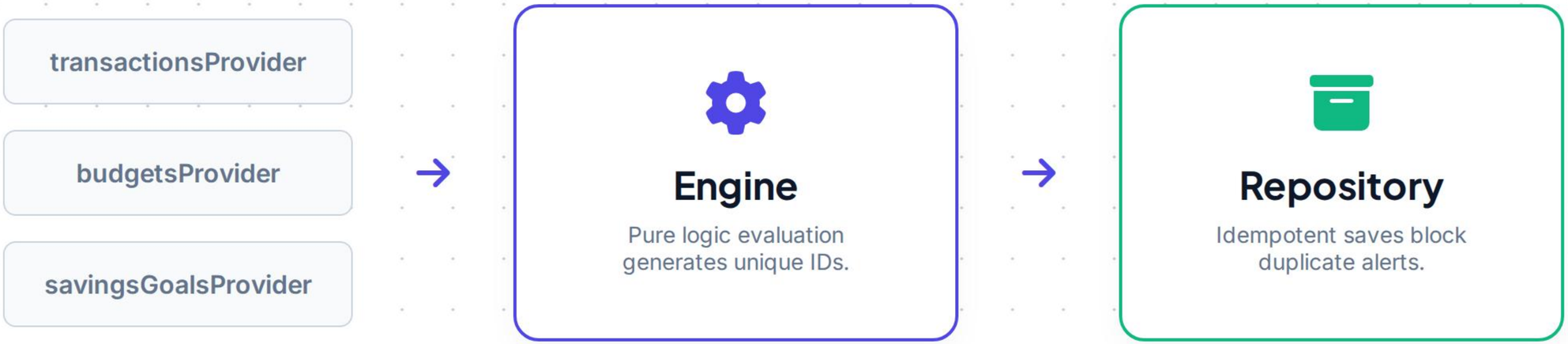
10 Bundled Lessons
Curated topics ready for offline reading.

Cloud Overrides
Firestore entries replace local content.

Progress Tracking
Read state synced to user profile.

Weekly Nudges
Automated study reminders triggered.

Deterministic Notification Engine



Administrative Shell

Dashboard
System stat aggregation.

Users List
Searchable student directory.

Support Ops
Query moderation UI.

CMS Content
Firestore lesson editor.

Design System

Theme Tokens: Standardized spacing and dimensions.

Typography: Plus Jakarta Sans & Inter.

Responsive: Adaptive constraints per breakpoint.



Header Element

Component body typography.



Deployment Ready

PennyPal architecture overview complete.

Image Sources



https://cdn.dribbble.com/userupload/48994892/file/82df82f56be42a584f9b990721038723.jpg?crop=0x2599-3500x5225&format=webp&resize=400x300&vertical=center

Source: dribbble.com



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